

1. The circuit used to store one bit of data is known as:
A. Register B. Encoder
C. Decoder D. Flip flop
2. In a microprocessor the clock signal is:
A. Is always generated internally
B. Is always supplied externally
C. May be generated internally or supplied externally
D. Is mostly supplied externally
3. Which of the following is the most popular family of SSI and MSI?
A. DTL B. TTL
C. ECL D. CMOS
4. In which memory the flip-flop is fabricated using standard P-N junction transistor?
A. Bipolar
B. Static MOS
C. Dynamic MOS
D. CMOS
5. Why are digital circuits easier to design than analog circuits?
A. They do not control electricity precisely over a wide range
B. They are made in the form of ICs
C. All elements of digital circuits are from the same family
D. They are smaller in size
6. If largest n -bit binary number requires ' d ' digits in decimal representation, then which of the following relation is correct?
A. $n = 2d$ B. $d = 2^n$
C. $d > n \log_2 10$ D. $d < n \log_2 10$
7. What is the maximum counting speed of a 4-bit binary counter which is composed of flip-flops with a propagation delay of 25 ns?
A. 1 MHz
B. 10 MHz
C. 100 MHz
D. 4 MHz
8. Structured design methodology is an approach to design that adheres to rules based on principal such as
A. Bottom-up design
B. Top-down refinement
C. Data flow analysis
D. All of the above
9. Which of the following model stipulates that the requirements be completely specified before the rest of the development can processed?
A. Waterfall
B. Rapid Application Development (RAD)
C. Iterative Development
D. Incremental Development
10. Black box testing is sometimes called
A. Data flow testing
B. Loop Testing
C. Behavioral Testing
D. Graph Based Testing
11. What are the signs that a software project is in trouble?
A. The product scope is poorly defined
B. Deadlines are unrealistic
C. Changes are managed poorly
D. All of the above

12. Which of these software characteristics are used to determine the scope of software project?
 - A. Only performance
 - B. Only context
 - C. Information objectives, function, performance
 - D. None of the above
13. Which of these software engineering activities are not a part of software processes?
 - A. Software dependence
 - B. Software development
 - C. Software validation
 - D. Software specification
14. Rapid application Development (RAD) model has
 - A. 2 phases B. 3 phases
 - C. 4 phases D. 5 phases
15. CMM stands for
 - A. Capability management module
 - B. Conservative maturity model
 - C. Capability maturity module
 - D. None of the above
16. contains information that defines valid values that are stored in a column or data type
 - A. View B. Rule
 - C. Index D. Default
17. Goals for the design of the logical schema includes?
 - A. Avoiding data inconsistency
 - B. Being able to construct queries easily
 - C. Being able to access data efficiently
 - D. All of the above
18. A transparent DBMS?
 - A. Cannot hide sensitive information from users
 - B. Keeps its logical structure hidden from users
 - C. Keeps its physical structure hidden from users
 - D. All of the above
19. A logical Schema?
 - A. Is the entire database
 - B. Is a standard way of organizing information into accessible parts?
 - C. Describes how data is actually stored on disk
 - D. All of the above
20. Embedded pointer provides.
 - A. A secondary access path
 - B. A physical record key
 - C. An inverted index
 - D. None of the above
21. In a Hierarchical model records are organized as:
 - A. Graph B. List
 - C. Links D. Tree
22. Which of the following is not a part of the operating system?
 - A. Supervisor
 - B. Performance monitor
 - C. Job control program
 - D. Input/output control program
23. Spooling is most beneficial in multiprogramming environment where
 - A. Most jobs are CPU- bound
 - B. Most jobs are I/O bound
 - C. Jobs are evenly divided as I/O bound and CPU bound
 - D. There is limited primary memory and need for secondary memory

24. FIFO scheduling is:
- Preemptive scheduling
 - Non-preemptive scheduling
 - Deadline scheduling
 - Fair-share scheduling
25. Trap is a/an
- Synchronous interrupt
 - Asynchronous interrupt
 - Hardware interrupt
 - Operating systems interrupt
26. Round robin is a
- Kind of magnetic drum
 - Process scheduling policy
 - Process synchronization policy
 - Memory allocation policy
27. A relation R is defined in $N \times N$, such that $(a,b) R (c,d)$ if $a+d = b+c$, the relation R is
- Reflexive but not transitive
 - Reflexive and transitive but not symmetric
 - An equivalence relation
 - A partial order
28. Regular grammar is
- Non-context free grammar
 - Context free grammar
 - English grammar
 - Context-sensitive a grammar
29. Which is not the correct statement?
- The class of regular set is closed under homeomorphisms
 - The class of regular sets is not closed under inverse homeomorphisms
 - The class of regular sets is closed under inverse homeomorphisms
 - The class of regular sets is closed under substitution
30. If a language is denoted by regular expression $L = (x)^*(x/yx)$, then which of the following is not a legal string within L
- yx
 - xyx
 - x
 - $xyxyx$
31. A grammar that produces more than one parse tree for some sentence is called
- Ambiguous
 - Unambiguous
 - Regular
 - None of above
32. Which programming use symbolic codes called MNEMONICS for developing the program?
- Machine Language
 - Low level language
 - High level language
 - Assembly language
33. Which program combine two or more object modules into a single object module or into an executable file.
- Linker
 - Interpreter
 - Assembler
 - Compiler
34. The process of assigning load addresses to the various parts of the program and adjusting the code and data in the program to reflect the assigned addresses is called
- Assembly
 - Parsing
 - Relocation
 - Symbol resolution
35. Which of the following statements is false?
- An unambiguous grammar has same leftmost and rightmost derivation
 - An LL (1) parser is top-down parser
 - LALR is more powerful than SLR

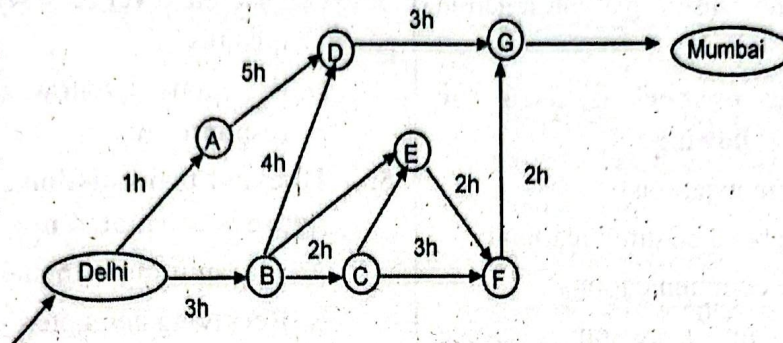
- D. An ambiguous following derivation does a top-down parser use while parsing an input string? The input is assumed to be scanned in left to right order.
36. Which of the following derivations does a top down parser use while parsing an input string? The input is assumed to be scanned in left to right order
- Leftmost derivation
 - Leftmost derivation traced out in reverse
 - Rightmost derivation
 - Rightmost derivation traced out in reverse
37. Database graphs can be produced as
- Bar graphs
 - Pie graphs
 - Line graphs
 - All of the above
38. Which of the following is not considered an input device for creating manipulating or reading images into interactive graphics programmed?
- Light pens
 - Digitizing tablets
 - RAM memory
 - Image digitizers
39. Computer aided design (CAD) programs are usually used to develop
- Business graphic
 - Graphics for specialized area such as engineering
 - Tutorial programs for school
 - All of the above
40. The amount of current that flows in short circuit is
- Zero
 - Very Low
 - Normal
 - Infinite
41. If a 100 Watts Bulb is ON for 10 hours, then what will be the amount of consumed Electricity?
- 100 watts
 - 100 watts per hour
 - 1000 watts (1KW)
 - 1 KWH=1 unit
42. If secondary numbers of turns in a transformer increases, the secondary output voltage of the transformer
- Increases
 - Decreases
 - Remains same
 - None of above
43. An RC differentiator acts as a
- Band pass filter
 - High pass filter
 - Low pass filter
 - AC open
44. The capacitor that produces an ac ground is called
- Coupling capacitor
 - DC open
 - Bypass capacitor
 - AC open
45. In "n" type material, majority carriers would be
- Holes
 - Dopants
 - Protons
 - Electrons
46. The primary function of the oscillator in an isolation amplifier is to
- Convert dc to high frequency ac
 - Convert dc to low frequency ac
 - Rectify high frequency ac to dc
 - Produce dual polarity dc voltages for the input to the demodulator
47. In a communication system, noise is most likely to affect the signal
- At the transmitter
 - In the channel
 - In the information source
 - At the destination

48. Indicate the false statement. Modulation is used to
- Reduce the bandwidth used
 - Separate differing transmission
 - Ensure that intelligence may be transmitted over long distance
 - Allow the use of practicable antennas
49. The acoustic channel is used for which of the following?
- UHF communications
 - Single sideband communications
 - Television communications
 - Person to person voice communication
50. Amplitude modulation is the process of
- Superimposing a low frequency on a high frequency
 - Superimposing a high frequency on a low frequency
 - Carrier interruption
 - Frequency shift and phase shift
51. Indicate the true statement. Most receivers conform to the
- Amplitude-modulated group
 - Frequency-modulated group
 - Super heterodyne group
 - Tuned radio frequency receiver group
52. A device that convert digital signals to analog signals is a/an
- Packet
 - Modem
 - Emulator
 - None of above
53. Which type of signal is represented by discrete values?
- Nonlinear
 - Noisy signal
 - Analog
 - Digital
54. Which of the following characterizes an analog quantity?
- It has a continuous set of values over a given range
 - It can be described with a finite number of steps
 - Discrete level represents change in a quantity
 - Its values follow a logarithmic response curve
55. To avoid transmission errors, a check figure is calculated by:
- Transmitting computer
 - Receiving computer
 - A personal computer
 - All of the above
56. The maximum data rate that a medium of 10 KHz and signal to noise ratio (SNR) of 30 dB is roughly
- 10 kbps
 - 30 kbps
 - 100 kbps
 - 3 kbps
57. If eraser is called box, 'box' is called 'pencil', 'pencil' is called 'sharpener', and 'sharpener' is called 'bag' what will a child write with?
- Eraser
 - Bag
 - Pencil
 - Sharpener
58. Find the nearest relationship of cricket: Pitch
- Ship: Dock
 - Boat: Harbour
 - Boxing: Ring
 - Wrestling: Track
59. Find the odd one out
- Epicenter
 - Seismology
 - Crater
 - Richter Scale

60. Pointing towards a person, a man said to a woman, " His mother is the only daughter of your father," How is the woman related to that person?

- A. Sister B. Daughter
C. Mother D. Wife

61. Which is the shortest route one can take to travel from Delhi to Mumbai?



- A. Delhi BDG Mumbai
C. Delhi BCFG Mumbai

- B. Delhi BEFG Mumbai
D. Delhi ADG Mumbai

62. Read the following passage and find the answer of the question: Who joined on Wednesday?

Six persons Amar, Bijaya, Chetan, Disha, Esha and Farukh took up a job with an organization in a week from Monday to Saturday, each of them joined for different posts on different days. The posts were of clerk, officer, technician, manager, supervisor and sales executives though not necessarily in the same order. Farukh joined as a manager on the first day. Bijay joined as a supervisor but neither on Wednesday nor on Friday. Disha joined as technical on Thursday. Chetan joined the firm as an officer before Disha and after Esha. Esha joined as a clerk on Tuesday. Amar joined as a sales executive.

- A. Chetan B. Clerk
C. Amar D. Manager

63. The sum of first 30 natural numbers is:

- A. 470 B. 468
C. 465 D. 463

64. The simple interest on Rs. 5200 for 2 years at 6% per annum is

- A. Rs 450
B. Rs. 524
C. Rs. 624
D. Rs. 324

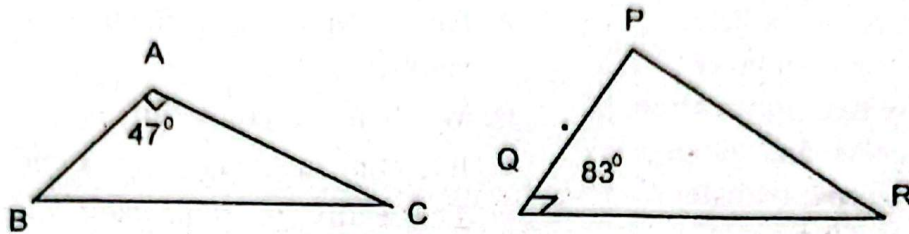
65. 1,14 expressed as a percent of 1.9 is

- A. 6 %
B. 60 %
C. 10 %
D. 90 %

66. A 3×2 matrix has

- A. 3 rows and 2 columns
B. 2 row and 3 columns
C. 1 row and 1 column
D. 5 rows and 5 columns

67. If ABC and PQR are similar triangles in which $\angle A = 47^\circ$ and $\angle Q = 83^\circ$, then $\angle C$ is:



- A. 50° B. 60°
C. 70° D. 80°
68. When 0.25 is converted into fraction, the result is
A. $1/2$ B. $1/3$
C. $1/4$ D. $1/5$
69. The square root of $(7+3\sqrt{5})(7-3\sqrt{5})$ is
A. 7 B. 2
C. 3 D. $3\sqrt{5}$
70. What is the angle made by the hour and minute hand of a watch at 12:00 noon?
A. 90° B. 30°
C. 0° D. 180°
71. In 15 minutes, a train running at the speed of 40 km/hr. covers distance
A. 15 km B. 40 km
C. 10 km D. 5 km
72. Which of the following is not a leap year?
A. 700 B. 800
C. 1200 D. 2000
73. The most important reason for failure of MIS is:
A. Use of improper tools for design
B. Noninvolvement of end-user
C. Improper specification
D. None of the above
74. A person machine-system and a highly integrated grouping of information-processing functions designed to provide management with a comprehensive picture of specific operation is called.
A. DSS B. MIS
C. IIS D. All of the above
75. Which of the following are areas of concern in the design model?
A. Architecture B. Data
C. Interfaces D. All of the above
76. Frames from one LAN can be transmitted to another LAN via the device
A. Router B. Bridge
C. Repeater D. Modem
77. What is the default subnet mask for a class C network?
A. 127.0.0.1 B. 255.0.0.0
C. 255.255.0.0 D. 255.255.255.0
78. Which of the following is false for CRC?
A. CRC is suited for detecting burst error
B. It is a non-secure hash function
C. CRC cannot work properly for detecting maliciously introduced errors
D. None of the above

79. Which of the following layer is not present in OSI model?
A. Physical Layer B. Network layer
C. Transport layer D. Internet layer
80. Which of the following application allows a user to access and change remote files without actual transfer?
A. DNS B. FTP
C. Telnet D. NFS
81. The data unit in the TCP/IP layer is called
A. Message B. Datagram
C. Frame D. Segment
82. When a class serves as base class for many derived classes, the situation is called:
A. Polymorphism
B. Hierarchical inheritance
C. Hybrid inheritance
D. Multipath inheritance
83. A constructor is called static constructor if
A. A constructor of a class is marked with static modifier
B. A constructor contains constant declaration
C. If the constructor can be triggered while creating instances of that class
D. None of the above
84. Singleton is an example of
A. Class initialization
B. Class inheritance
C. Object oriented design
D. Object oriented framework
85. A template can be used to create a family of?
A. Classes
B. Functions
C. Class and function
D. None of above
86. Which is not true about "HAS-A" relationship?
A. One entity is contained within another
B. Whole part relationship
C. It is implemented using "extends"
D. The entity in right hand side of "Has-A " is defined as attribute of class in the left-hand side of "HAS-A"
87. Which of the following cohesion deals with "execution at some time"?
A. Procedural cohesion
B. Temporal cohesion
C. Logical cohesion
D. Time stamp cohesion
88. Decision support programs are designed to help managers make:
A. Budget projection
B. Visual presentation
C. Business decision
D. Vacation schedules
89. Programming a robot by physically moving it through the trajectory you want it to follow is called
A. Contact sensing control
B. Continuous path control
C. Robot vision control
D. Pick and place control
90. An artificial intelligence technique that allows computer to understand associations and relationships between objects and events is called.
A. Heuristic processing
B. Cognitive science
C. Relative symbolism
D. Pattern matching

91. Which of the following ways is a pre-order traversal?
- Root → left sub tree → right sub tree
 - Root → Right sub tree → left sub tree
 - Right sub tree → left sub tree → Root
 - Left sub tree → right sub tree → Root
92. Which of the following algorithm design technique is used the quick sort algorithm?
- Dynamic programming
 - Backtracking
 - Divide and conquer
 - Greedy method
93. Which statement is true for a B-Tree?
- All entities of a node are greater than or equal to the entries in the node's children
 - All leaves are at the exact same depth
 - All nodes contain the exact same number of children
 - All non-leaf nodes have exact same number of children
94. If time taken to sort 100 entries by quick sort is 10 seconds, what will be taken to sort 1000 entries?
- 100
 - 150
 - 1000
 - 4000
95. The complexity of Binary search algorithm is
- $O(n)$
 - $O(\log n)$
 - $O(1)$
 - $O(n \log n)$

96. Which of the following is/are true?
- Spanning tree is unique for a graph
 - Prim's and Kruskal's algorithm can give same spanning trees
- 1 only
 - 2 only
 - Both 1 and 2
 - None of the above
97. Memory address refers to the successive memory words and the machine is called as:
- Word addressable
 - Byte addressable
 - Bit addressable
 - Terra bytes addressable
98. The register that keeps track of the instructions in the machine is called as:
- Control registers
 - Program counter
 - Status register
 - Direct register
99. The communication between central system and the outside environment is done by:
- Input-output subsystem
 - Control system
 - Memory system
 - Logical system
100. Cache memory words on the principle of
- Locality of data
 - Locality of memory
 - Locality of reference
 - Locality of reference & memory

ANSWER SHEET

1.D	2.C	3.D	4.A	5.A	6.C	7.D	8.C	9.A	10.C
11.D	12.C	13.A	14.C	15.C	16.C	17.D	18.C	19.B	20.A
21.D	22.D	23.C	24.B	25.C	26.B	27.C	28.A	29.B	30.D
31.A	32.D	33.A	34.C	35.A	36.A	37.D	38.C	39.B	40.A
41.C	42.C	43.B	44.C	45.D	46.A	47.B	48.A	49.D	50.B
51.C	52.B	53.D	54.A	55.B	56.B	57.D	58.C	59.C	60.C
61.D	62.A	63.C	64.C	65.B	66.A	67.A	68.C	69.B	70.C
71.C	72.A	73.B	74.B	75.D	76.B	77.D	78.A	79.D	80.C
81.C	82.D	83.D	84.C	85.B	86.D	87.B	88.C	89.B	90.D
91.A	92.C	93.B	94.A	95.B	96.B	97.A	98.B	99.A	100.C